Lesson 5-5 · Period 1 (Quick)

Function Operations + Composition Order DOK-3 (Standalone)

Klimsara-adapted · Student-centered · No Blooket

Do Now — Profit as a function operation

DOK 1 | 5 min | *bridge to function ops*

Model & Discuss — Grooming USA Revenue: \$25 per pet. Cost: \$15 per pet + \$1,000/month.

1. Define $r(x)$ and $c(x)$.
2. Did they profit on 95 pets? On 110?
3. How could you write ONE function for profit?

Think alone 3 min — write $P(x) = r(x) - c(x)$ in your notes.

Launch — Function Operations Rules (Ex 1 + Ex 4) Algebra 2 · Lesson 5-5 · Period 1

DOK 2 | 12 min | *teacher models*

Function Operations Rules — keep on your page

1. $(f \pm g)(x) = f(x) \pm g(x)$; $(fg)(x) = f(x) \cdot g(x)$; $(f/g)(x) = f(x)/g(x)$, $g(x) \neq 0$.
2. Domain of combined function = intersection of domains, minus new excluded values.
3. Composition: $(f \circ g)(x) = f(g(x))$ — apply g FIRST, then f.
4. $f \circ g \neq g \circ f$ in general. **ORDER MATTERS.**

30-sec partner check Test $f(g(2))$ vs $g(f(2))$ with $f(x) = 2x - 1$ and $g(x) = 3x$.
Are the results the same?

Explore — TPS + DOK-3 Capstone

DOK 2-3 **33 min** *student work*

8 items in order. Plan ~12 min for Practice #32 (Music Store). 45-min Wed F: skip #34 + #28.

Try It 1 · Add/subtract functions $f(x) = 2x^2 + 7x - 1$, $g(x) = 3 - 2x$. Find $f + g$ and $f - g$.

Try It 4 · Compose functions $f(x) = 2x - 1$, $g(x) = 3x$. Find $f(g(2))$ and $f(g(x))$.

#20 / #21 · Operations; #24 · Domain; #28 / #34 · Compose + mixed See packet. Work in order. Domain intersections matter for #24.

Share / Summary — Order Matters

DOK 2 | 5 min | *DOK-3 reveal*

Sentence frame · Practice #32 Music Store "Applying ________ first gives ________, then ________ gives ________.
If I reverse the order I get ________, so $(f \circ g)(x) \neq (g \circ f)(x)$."

One pair presents — class signals thumbs.

Lesson Wrap Composition order matters in real decisions: discounts, unit conversions, multi-step processes.
$f \circ g \neq g \circ f$ in general — always verify with a test value.

Exit Ticket — Summary Recap

Algebra 2 · Lesson 5-5 · Period 1

DOK 1-2 | 3 min | *not graded*

Complete on your exit ticket

1. $f \circ g$ means ______________________ is applied ___________.
2. An example where $f \circ g \neq g \circ f$: ______________________.
3. For a double discount, the better order for the consumer is ___________ because _________________.
4. One biggest thing I learned today: ____________________________.